Wireshark Lab: DNS v8.0

Supplement to *Computer Networking: A Top-Down Approach*, 8th ed., J.F. Kurose and K.W. Ross

"Tell me and I forget. Show me and I remember. Involve me and I understand."
Chinese proverb

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As described in Section 2.4 of the text,[1] the Domain Name System (DNS) translates hostnames to IP addresses, fulfilling a critical role in the Internet infrastructure. In this lab, we will take a closer look at the client side of DNS. Recall that the client's role in the DNS is relatively simple — a client sends a *query* to its local DNS server, and receives a *response* back. As shown in Figures 2.19 and 2.20 in the textbook, much can go on "under the covers," invisible to the DNS clients, as the hierarchical DNS servers communicate with each other to either recursively or iteratively resolve the client's DNS query. From the DNS client's standpoint, however, the protocol is quite simple — a query is formulated to the local DNS server and a response is received from that server.

Before beginning this lab, you will probably want to review DNS by reading Section 2.4 of the text. In particular, you may want to review the material on **local DNS servers**, **DNS caching**, **DNS records and messages**, and the **TYPE field** in the DNS record.

1. nslookup

In this lab, we will make extensive use of the *nslookup* tool, which is available in most Linux/Unix and Microsoft platforms today. To run *nslookup* in Linux/Unix, simply type the `nslookup` command on the command line. To run it in Windows, open the Command Prompt and run `nslookup` on the command line.

In its most basic operation, *nslookup* allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server (see the textbook for definitions of these terms). To accomplish this task, *nslookup* sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

The screenshot below shows the results of three independent *nslookup* commands (displayed in the Windows Command Prompt). In this example, the client host is located on the campus of Polytechnic University in Brooklyn, where the default local DNS server is `dns-prime.poly.edu`. When running *nslookup*, if no DNS server is specified, then *nslookup* sends the query to the default DNS server (`dns-prime.poly.edu`). Consider the first command:

```
nslookup www.mit.edu
```

In words, this command is saying "please send me the IP address for the host `www.mit.edu`".

[1] References to figures and sections are for the 8th edition of our text, *Computer Networks, A Top-down Approach, 8th ed.*, J.F. Kurose and K.W. Ross, Addison-Wesley/Pearson, 2020.

As shown in the screenshot, the response provides two pieces of information: (1) the name and IP address of the DNS server that provides the answer; and (2) the answer itself, which is the hostname and IP address of `www.mit.edu`. Although the response came from the local DNS server at Polytechnic University, it is quite possible that this local DNS server iteratively contacted several other DNS servers to get the answer, as described in Section 2.4 of the textbook.

Now consider the second command:

```
nslookup -type=NS mit.edu
```

In this example, we have provided the option `-type=NS` and the domain `mit.edu`. This causes *nslookup* to send a query for a type-NS record to the default local DNS server. In words, the query is saying, "please send me the hostnames of the authoritative DNS servers for `mit.edu`". (When the `-type` option is not used, *nslookup* uses the default, which is to query for type-A records.) The answer first indicates the DNS server providing the answer (the default local DNS server) along with three MIT nameservers. Each of these servers is indeed an authoritative DNS server for hosts on the MIT campus. However, *nslookup* also indicates that the answer is "non-authoritative," meaning that this answer came from the cache of some server rather than directly from an authoritative MIT DNS server. The answer also includes the IP addresses of the authoritative DNS servers at MIT.

Now finally consider the third command:

```
nslookup www.aiit.or.kr bitsy.mit.edu
```

In this example, we indicate that we want the query sent to the DNS server `bitsy.mit.edu` rather than to the default DNS server (`dns-prime.poly.edu`). Thus, the query and reply transaction takes place directly between our querying host and `bitsy.mit.edu`. In this example, the DNS server `bitsy.mit.edu` provides the IP address of the host `www.aiit.or.kr`, which is a web server at the Advanced Institute of Information Technology (in Korea).

The general syntax of *nslookup* commands is:

```
nslookup -option1 -option2 host-to-find dns-server
```

In general, *nslookup* can be run with zero, one, two or more options. As we have seen in the examples above, the `dns-server` argument is optional as well; if it is not supplied, the query is sent to the default local DNS server.

Now that we have provided an overview of *nslookup*, it is time to test drive it yourself. Do the following (and write down the results):

Q1. Run *nslookup* to obtain the IP address of a Web server in Asia. What is the IP address of that server?

```
root@DESKTOP-SB5GJ2U:~# nslookup www.vnexpress.net
Server:         203.113.188.1
Address:        203.113.188.1#53

Non-authoritative answer:
www.vnexpress.net       canonical name = vnexpress.net.
Name:   vnexpress.net
Address: 111.65.250.2
Name:   vnexpress.net
Address: 2001:df0:66:40::16
```

ANS: 111.65.250.2

Q2. Run *nslookup* to determine the authoritative DNS servers for a university in Europe.

```
root@DESKTOP-SB5GJ2U:~# nslookup -type=NS ox.ac.uk
Server:         203.113.188.1
Address:        203.113.188.1#53

Non-authoritative answer:
ox.ac.uk        nameserver = dns2.ox.ac.uk.
ox.ac.uk        nameserver = auth4.dns.ox.ac.uk.
ox.ac.uk        nameserver = dns1.ox.ac.uk.
ox.ac.uk        nameserver = ns2.ja.net.
ox.ac.uk        nameserver = dns0.ox.ac.uk.
ox.ac.uk        nameserver = auth5.dns.ox.ac.uk.
ox.ac.uk        nameserver = auth6.dns.ox.ac.uk.

Authoritative answers can be found from:
dns0.ox.ac.uk   internet address = 129.67.1.190
auth5.dns.ox.ac.uk      internet address = 93.93.128.67
dns1.ox.ac.uk   internet address = 129.67.1.191
auth4.dns.ox.ac.uk      internet address = 45.33.127.156
auth6.dns.ox.ac.uk      internet address = 185.24.221.32
ns2.ja.net      internet address = 193.63.105.17
dns2.ox.ac.uk   internet address = 163.1.2.190
auth5.dns.ox.ac.uk      has AAAA address 2a00:1098:0:80:1000::10
auth4.dns.ox.ac.uk      has AAAA address 2600:3c00::f03c:91ff:fe96:beac
auth6.dns.ox.ac.uk      has AAAA address 2a02:2770:11:0:21a:4aff:febe:759b
ns2.ja.net      has AAAA address 2001:630:0:45::11

root@DESKTOP-SB5GJ2U:~#
```

ANS: There are multiple authoritative servers. To confirm the authoritative DNS servers, we perform the same DNS query of one of the servers that can provide authoritative answers.

```
root@DESKTOP-SB5GJ2U:~# nslookup -type=NS ox.ac.uk dns0.ox.ac.uk
Server:         dns0.ox.ac.uk
Address:        129.67.1.190#53

ox.ac.uk        nameserver = ns2.ja.net.
ox.ac.uk        nameserver = auth6.dns.ox.ac.uk.
ox.ac.uk        nameserver = dns0.ox.ac.uk.
ox.ac.uk        nameserver = auth5.dns.ox.ac.uk.
ox.ac.uk        nameserver = dns1.ox.ac.uk.
ox.ac.uk        nameserver = auth4.dns.ox.ac.uk.
ox.ac.uk        nameserver = dns2.ox.ac.uk.

root@DESKTOP-SB5GJ2U:~#
```

Q3. Run *nslookup* so that one of the DNS servers obtained in Question 2 is queried

for the mail servers for Yahoo! mail. What is its IP address?

```
root@DESKTOP-SB5GJ2U:~# nslookup mail.yahoo.com dns0.ox.ac.uk
Server:         dns0.ox.ac.uk
Address:        129.67.1.190#53

** server can't find mail.yahoo.com: REFUSED

root@DESKTOP-SB5GJ2U:~#
```

ANS: As you can see, I cannot get the DNS servers to answer a query for a Yahoo mail server.

2. ipconfig

ipconfig (for Windows) and *ifconfig* (for Linux/Unix) are among the most useful little utilities in your host, especially for debugging network issues. Here we will only describe *ipconfig*, although the Linux/Unix *ifconfig* is very similar. *ipconfig* can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on. For example, you can retrieve all this information about your host simply by entering

```
ipconfig /all
```

into the Command Prompt.

ipconfig is also very useful for managing the DNS information stored in your host. In Section 2.5 we learned that a host can cache DNS records it recently obtained. To see these cached records, after the prompt `C:\>` provide the following command:

```
ipconfig /displaydns
```

Each entry shows the remaining Time to Live (TTL) in seconds. To clear the cache, enter:

```
ipconfig /flushdns
```

Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

3. Tracing DNS with Wireshark

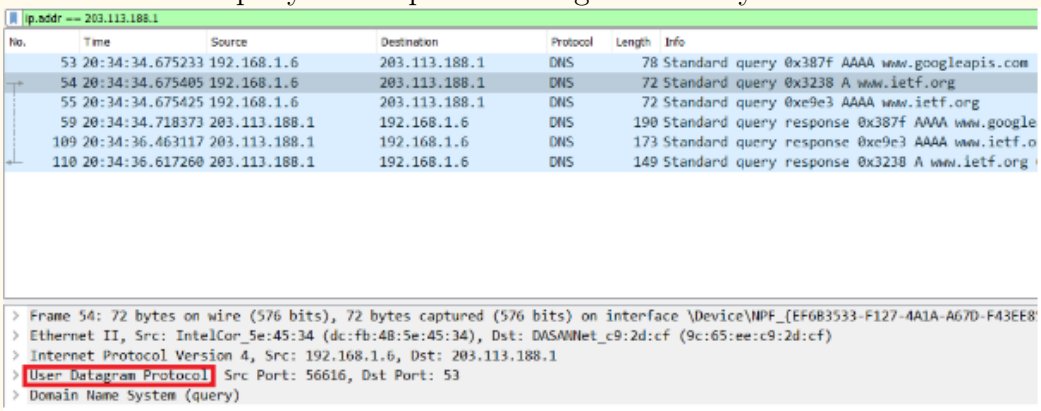
Now that we are familiar with *nslookup* and *ipconfig*, we are ready to get down to some serious business. Let us first capture the DNS packets that are generated by ordinary web-surfing activity.

- Use `ipconfig` to empty the DNS cache in your host.
- Open your browser and empty your browser cache. (With Internet Explorer, go to the *Tools* menu and select *Internet Options*; then in the *General* tab select *Delete Files*.)
- Open Wireshark and enter `ip.addr == your_IP_address` into the filter, where you obtain `your_IP_address` with `ipconfig`. This filter removes all packets that neither originate nor are destined to your host.
- Start packet capture in Wireshark.

- With your browser, visit the Web page: http://www.ietf.org
- Stop packet capture.

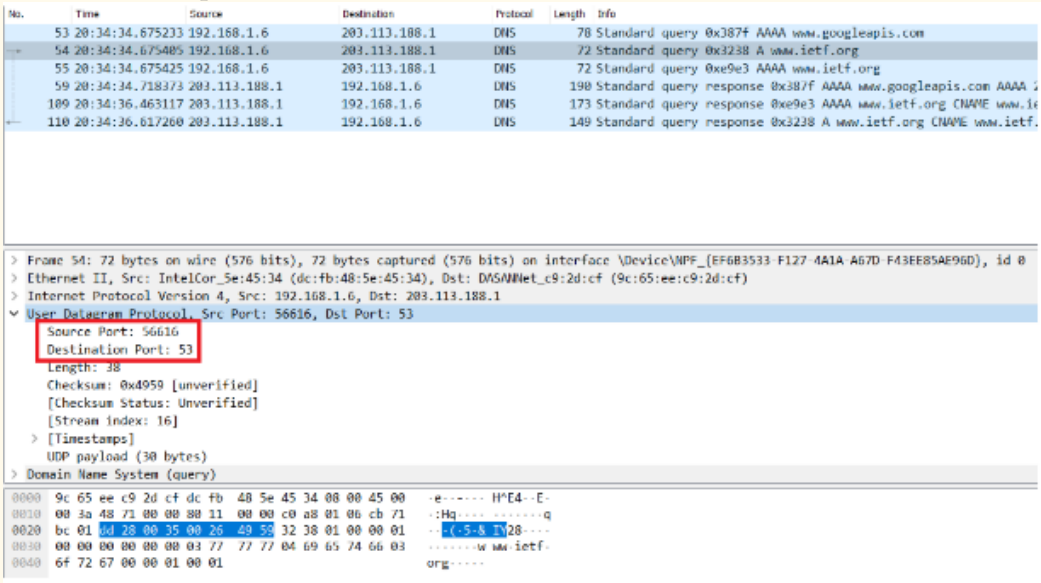
> **Note:** If you are unable to run Wireshark on a live network connection, you can download a packet trace file captured while following the steps above.[a] Whenever possible, when answering a question below, hand in a printout of the packet(s) within the trace that you used to answer the question and annotate the printout to explain your answer.
>
> [a] Download the zip file http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip and extract the file `dns-ethereal-trace-1`. Load it into Wireshark via *File* → *Open*.

Q4. Locate the DNS query and response messages. Are they sent over UDP or TCP?

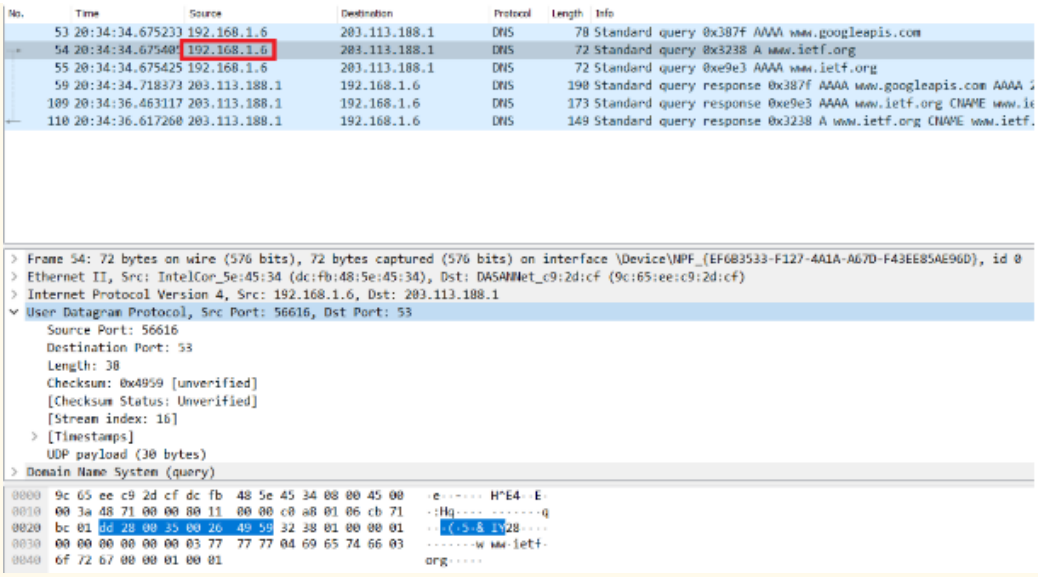


ANS: They are sent over UDP.

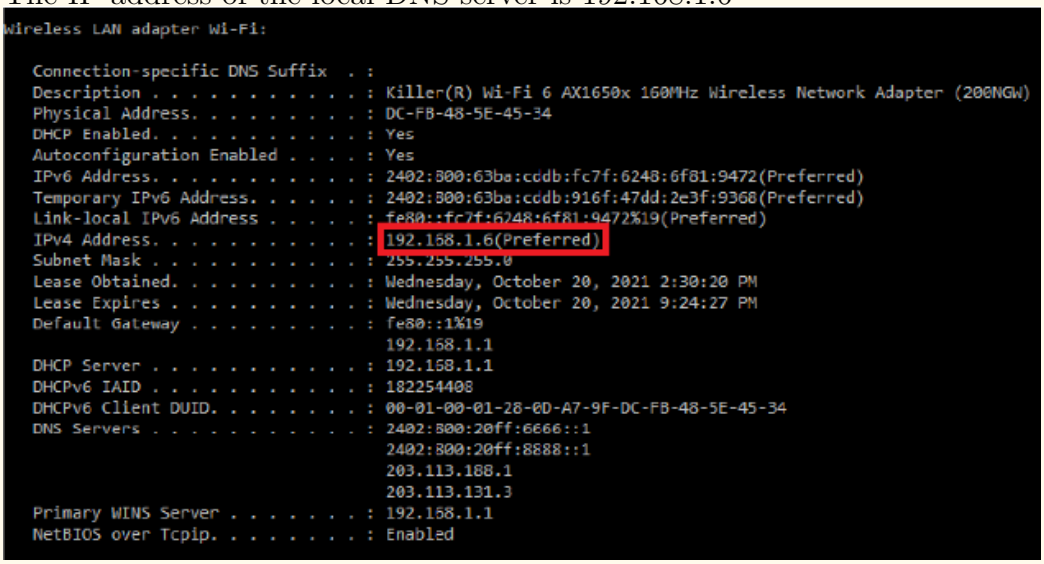
Q5. What is the destination port for the DNS query message? What is the source port of the DNS response message?
ANS: Source port: 56616



Q6. To what IP address is the DNS query message sent? Use `ipconfig` to determine the IP address of your local DNS server. Are these two IP addresses the same?
ANS: The DNS query message is sent to 192.168.1.6

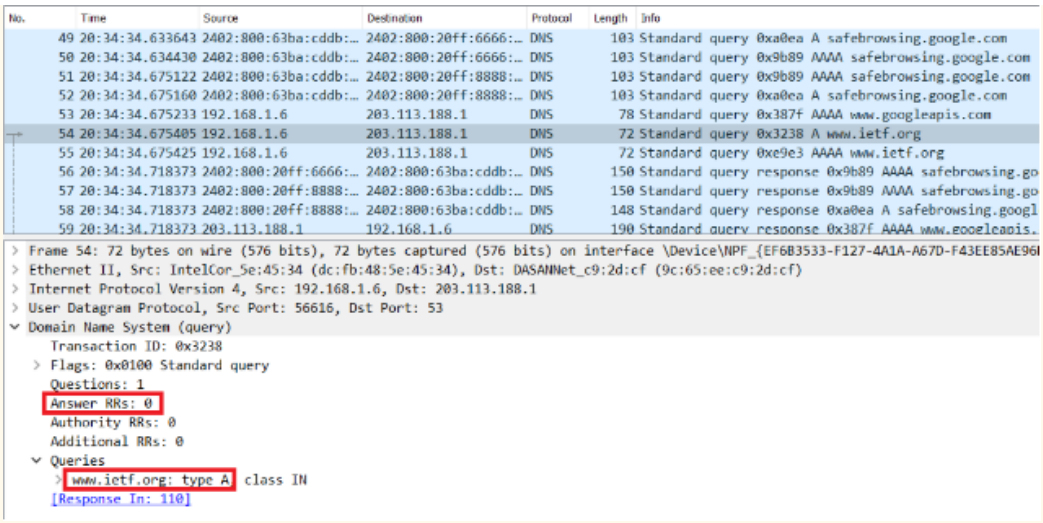


The IP address of the local DNS server is 192.168.1.6



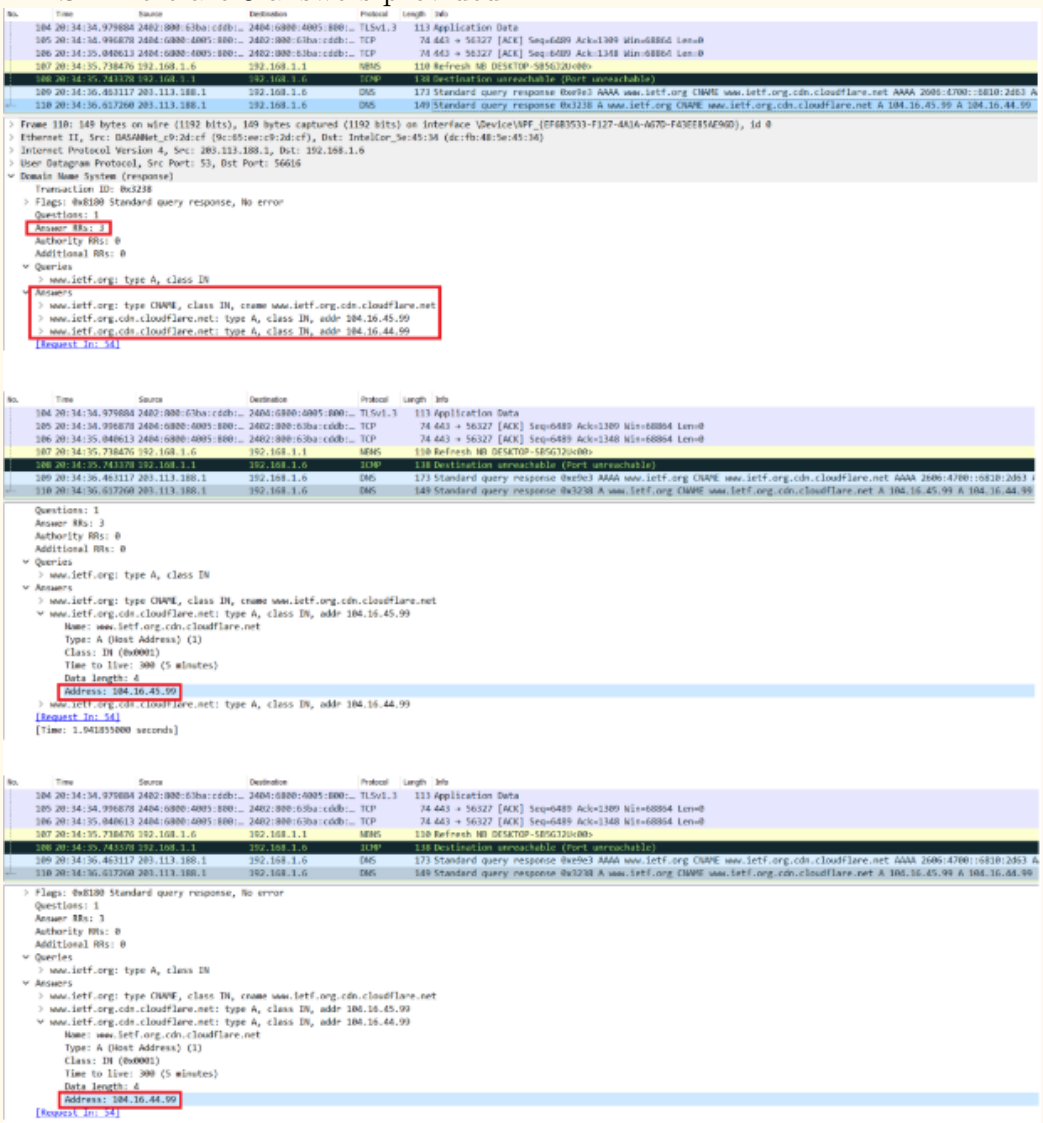
These two IP addresses are the same.

Q7. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?
ANS: The type of DNS query is A. The query message does not contain any answers.



Q8. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain?

ANS: There are 3 answers provided.



Q9. Consider the subsequent TCP SYN packet sent by your host. Does the des-

tination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?
ANS: The destination IP address of the SYN packet is 104.16.45.99, the same address that was provided in the DNS respones message as the type "A" address of the webpage.

Q10. This web page contains images. Before retrieving each image, does your host issue new DNS queries?
ANS: No because all the images are loaded from www.ietf.org so there is no need for additional DNS queries.

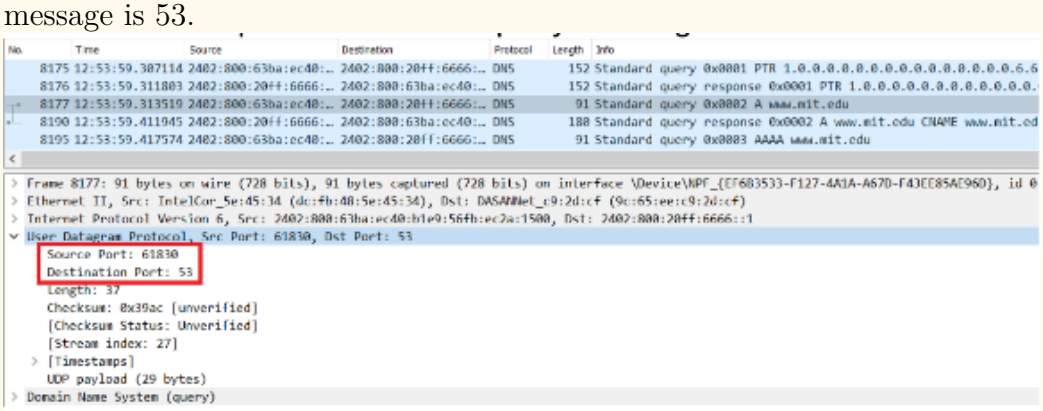
nslookup Experiment 1: Basic A Record Query

Now let us play with *nslookup*.[2]

- Start packet capture.
- Do an *nslookup* on `www.mit.edu`.
- Stop packet capture.

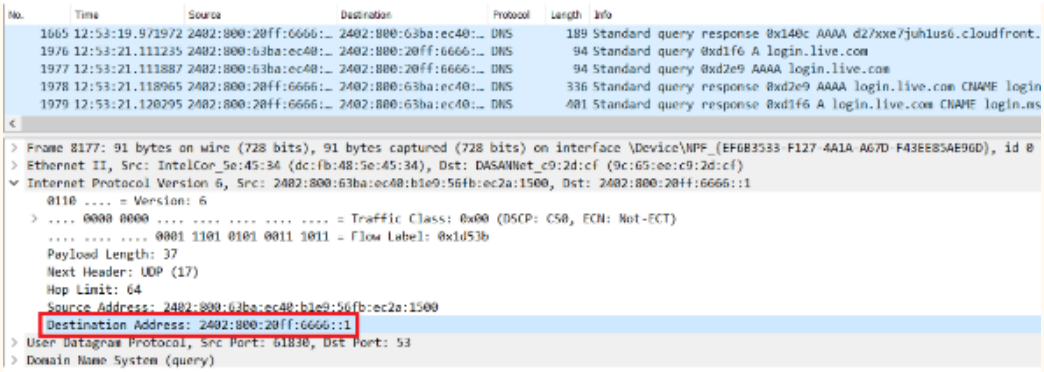
Note that *nslookup* actually sends three DNS queries and receives three DNS responses. For the purpose of this assignment, **ignore the first two sets of queries/responses**, as they are specific to *nslookup* and are not normally generated by standard Internet applications. Focus instead on the last query and response messages.

Q11. What is the destination port for the DNS query message? What is the source port of the DNS response message?
ANS: The source port response message is 61830. The destination port query message is 53.

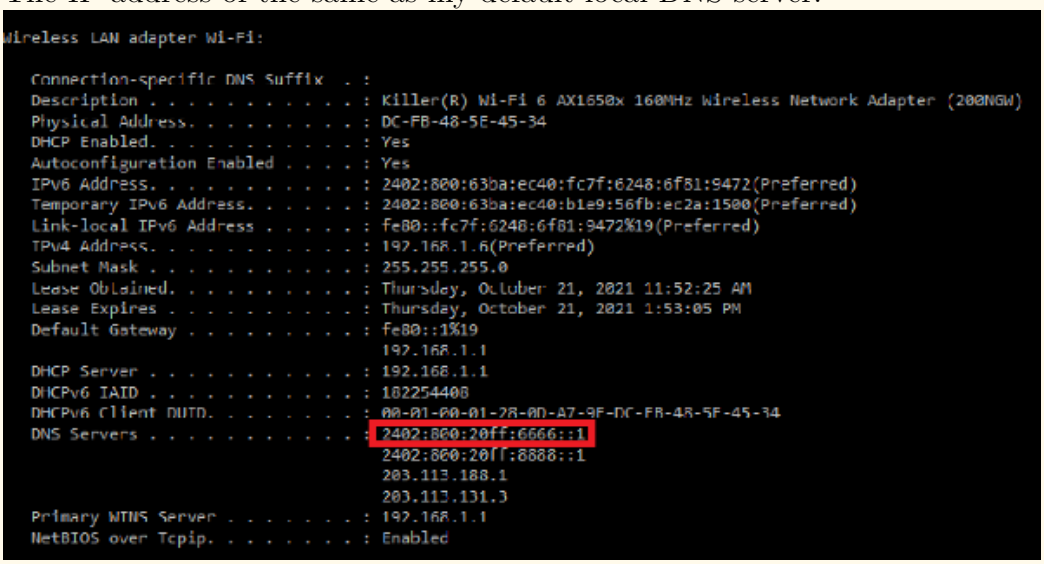


Q12. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?
ANS: The DNS query message is sent to 2402:800:20ff:6666::1.

[2] If you are unable to run Wireshark and capture a trace file, use the trace file `dns-ethereal-trace-2` in the zip file http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip.

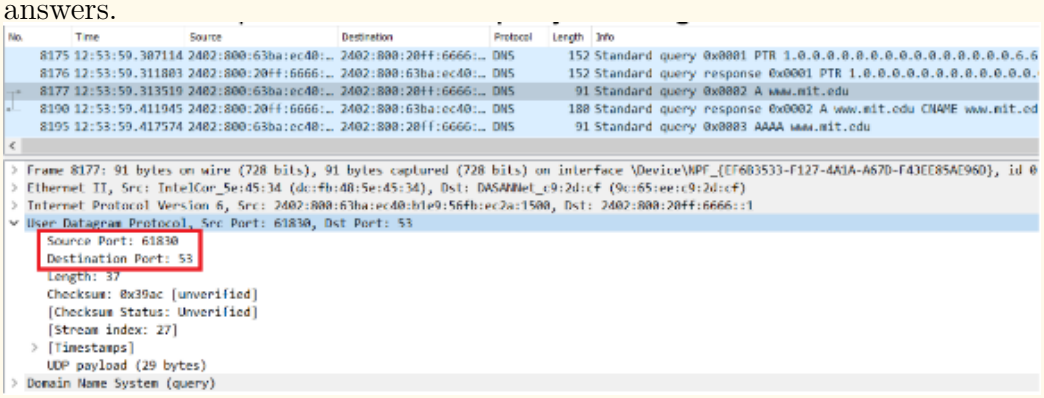


The IP address of the same as my default local DNS server.



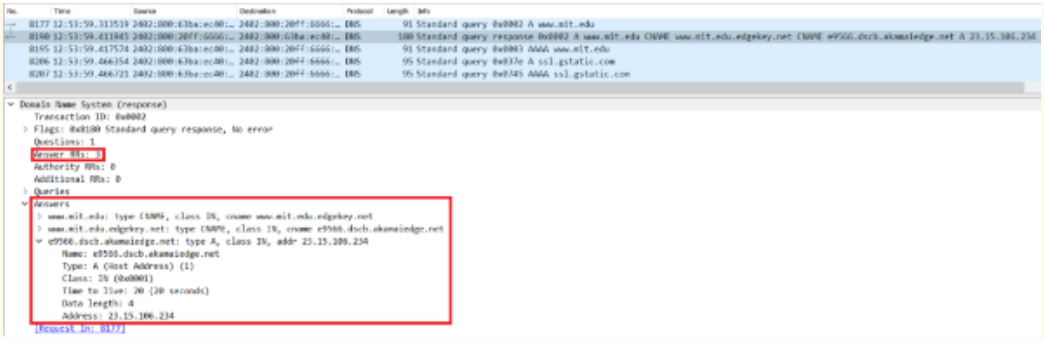
Q13. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

ANS: The type of DNS query is A. The query message does not contain any answers.



Q14. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain?

ANS: There are 3 answers provided.



Q15. Provide a screenshot.
ANS: Each question I have already provided a screenshot.

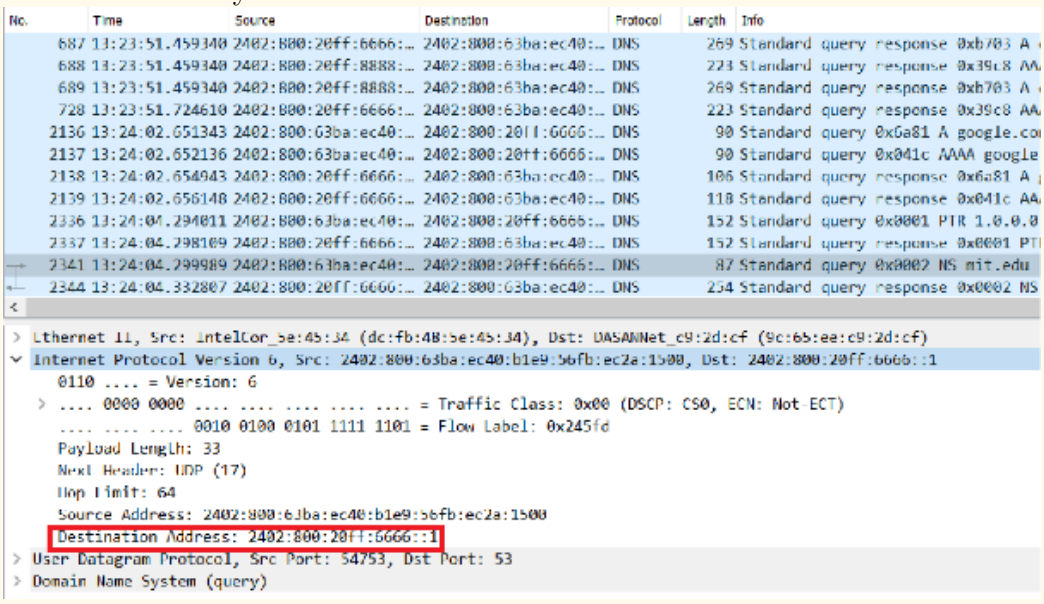
nslookup Experiment 2: NS Record Query

Now repeat the previous experiment, but instead issue the command:

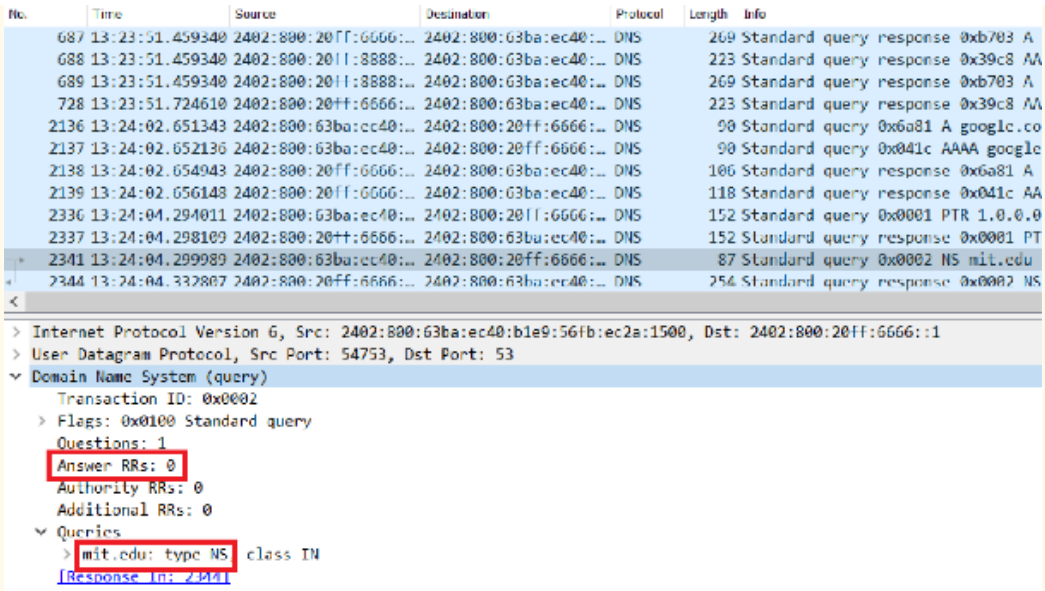
```
nslookup -type=NS mit.edu
```

> **Note:** If you are unable to run Wireshark and capture a trace file, use the trace file `dns-ethereal-trace-3` in the zip file http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip.

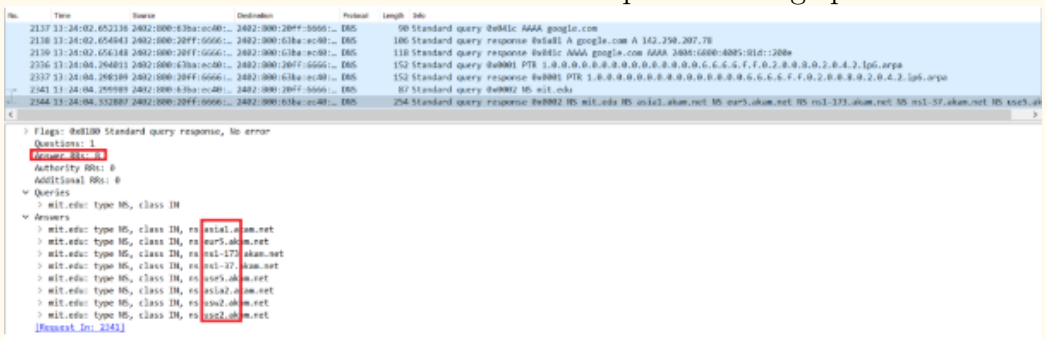
Q16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?
ANS: The IP address that the DNS query message sent to: 2402:800:20ff:6666::1. It the same as my default local DNS server.



Q17. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?
ANS: Type of DNS query: NS. The query message does not contain any answers.



Q18. Examine the DNS response message. What MIT nameservers does the response message provide? Does this response message also provide the IP addresses of the MIT nameservers?
ANS: There are 8 MIT nameservers that response message provides.



Q19. Provide a screenshot.
ANS: I have already provided screenshots for each question.

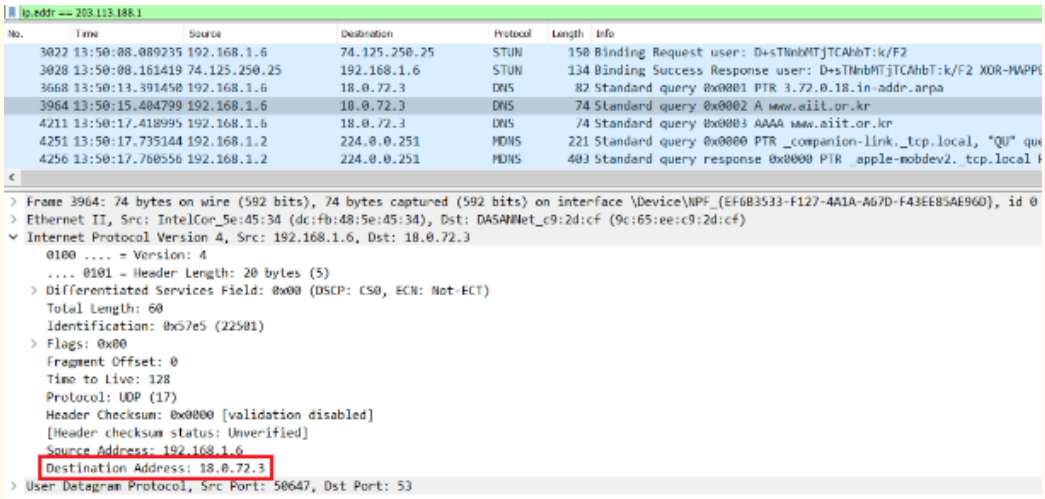
nslookup Experiment 3: Querying a Specific DNS Server

Now repeat the previous experiment, but instead issue the command:

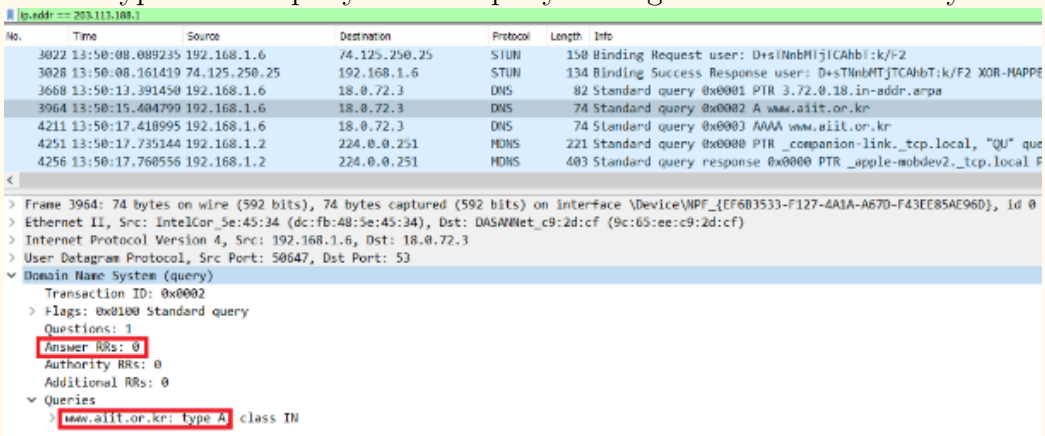
```
nslookup www.aiit.or.kr bitsy.mit.edu
```

> **Note:** If you are unable to run Wireshark and capture a trace file, use the trace file `dns-ethereal-trace-4` in the zip file http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip.

Q20. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? If not, what does the IP address correspond to?
ANS: The IP address that the DNS query message sent to: 18.0.72.3. This IP address corressponds to bitsy.mit.edu.



Q21. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?
ANS: Type of DNS query: A. The query message does not contain any answers.



Q22. Examine the DNS response message. How many "answers" are provided? What does each of these answers contain?
ANS: There is no response message.

Q23. Provide a screenshot.
ANS: I have already provided screenshots for each question.